MeloVOCOS: Decoupling Waveform Reconstruction for a Neural TTS

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Abstract

Neural text-to-speech systems based on VITS-like architectures jointly optimize text conditioning, acoustic representation, and waveform generation, making it difficult to isolate the effect of waveform synthesis. We propose MeloVOCOS, a MeloTTS architecture that preserves the main linguistic, duration, alignment, posterior-encoding, and flow-based components while assigning waveform reconstruction to a frozen, pretrained VOCOS neural synthesizer conditioned on log-mel representations. We compare MeloVOCOS with MeloTTS under matched conditions using intelligibility, speaker similarity, perceived speech and voice quality, and training behavior. MeloVOCOS reduces training time and the number of training steps by approximately $4.2\times$, maintaining a similar per-epoch cost. Results suggest that waveform reconstruction can be partially delegated to a pretrained synthesizer while preserving the model's ability to adapt its internal acoustic representations to the target speaker.

Index Terms: Text-to-Speech, Neural Speech Synthesis, MeloTTS, VOCOS, Waveform Reconstruction.

1. Introduction

Text-to-Speech (TTS) systems convert written text into speech waveforms through modular pipelines or highly integrated end-to-end (E2E) models. In modular systems, different functions are assigned to separate components, including linguistic processing, duration modelling, acoustic generation, and waveform synthesis. In contrast, E2E approaches learn these functions jointly, often improving synthesis quality but reducing interpretability and control over intermediate representations and the training process. Despite these architectural differences, recent advances have significantly improved the naturalness, intelligibility, and speaker similarity across architectures, enabling high-quality multilingual synthesis.

In VITS-like architectures, which integrate several modular components into a jointly trained pipeline, a significant portion of training is devoted to waveform reconstruction. While upstream components learn linguistic, temporal, and acoustic representations associated with speech and speakers, the synthesizer model must reproduce acoustic cues related to pronunciation, prosody, timing, and speaker identity from intermediate representations. Among these neural vocoders, HiFi-GAN has become one of the most widely adopted solutions. Architectures such as MeloTTS [1] integrate adversarial learning into the training pipeline, coupling acoustic generation with waveform reconstruction. MeloTTS has been adopted in several recent speech generation and multilingual synthesis pipelines [2, 3], particularly for real-time conversion within multilingual translation on a cloned speaker [4, 5, 6]. While this design supports high-fidelity synthesis, it requires additional components such as discriminator networks and adversarial losses. These dependencies increase optimization complexity and computational cost, suggesting that waveform reconstruction may be better decoupled for efficient training.

Neural vocoding has demonstrated that high-quality waveform generation can be achieved from pretrained models. VOCOS [7] adopts a frequency-domain reconstruction strategy from mel-spectrograms and can be deployed as a frozen synthesizer without retraining the synthesizer. This separation suggests that waveform reconstruction may not need to be relearned for every target speaker.

To investigate this question, we propose MeloVOCOS, a modified version of MeloTTS in which the original HiFi-GAN generator is replaced by a pretrained, frozen VOCOS model, eliminating the need to adapt the synthesizer itself. The proposed architecture retains the original linguistic encoder, attention mechanisms, deterministic and stochastic duration predictors, and normalizing flow components, while assigning waveform reconstruction to VOCOS. By delegating waveform reconstruction to a pretrained synthesizer, training can focus more directly on adapting the internal acoustic representations to the target speaker. Since VOCOS expects log-mel representations rather than the latent features used by the original HiFi-GAN module, the acoustic representation and training losses are accordingly adapted. This reduces the number of trainable objectives associated with waveform generation and simplifies the overall optimization process.

Experimental evaluation compares the original MeloTTS architecture and the proposed MeloVOCOS framework under identical conditions. The study analyses the impact of replacing adversarial waveform generation with pretrained synthesis on training efficiency, computational requirements, and synthesis performance. We report objective metrics for intelligibility, speaker identity, and perceived speech and voice quality. The main contributions of this work are as follows:

- a MeloTTS-derived architecture replacing the original HiFi-GAN generator with a frozen pretrained VOCOS synthesizer;
- an adaptation of the acoustic representation and training losses to the log-mel feature space expected by VOCOS;
- a simplified training strategy that reduces the optimization burden associated with waveform reconstruction;
- an open-source implementation of the proposed architecture to support reproducibility and future research, available in the project repository: `URL_hidden_for_anonymity`;
- A comparative evaluation of HiFi-GAN and VOCOS within the same MeloTTS framework, analysing efficiency and synthesis performance under equivalent conditions.

2. Related Work

Neural TTS systems typically comprise three stages: linguistic encoding, acoustic generation, and waveform synthesis. Early autoregressive architectures, such as Tacotron [8] and Tacotron2 [9], generated mel-spectrograms followed by neural vocoding. Although they improved speech naturalness, their autoregressive decoding limited inference speed, robustness, and controllability.

Subsequent non-autoregressive and flow-based architectures improved efficiency and stability through explicit duration modeling, parallel acoustic generation, and latent-variable formulations. FastSpeech [10] introduced explicit duration prediction for parallel generation, while VITS [11] unified variational inference, normalizing flows, stochastic duration prediction, and adversarial waveform generation within a single E2E framework. VITS2 further improved the architecture through training and adversarial refinements [12], whereas Matcha-TTS adopted conditional flow matching for efficient acoustic modelling [13]. Nevertheless, VITS-like architectures still jointly optimize acoustic representation learning and waveform generation.

Despite the evolution of E2E TTS systems, modularity remains a valuable design principle, allowing individual components to be analyzed and replaced without modifying the complete pipeline. MeloTTS [1], a hybrid modular and E2E design, combines grapheme-to-phoneme conversion, text encoding, duration prediction, posterior encoding, normalizing flows, and an adversarial waveform synthesizer within a unified VITS-based architecture. During inference (top of Fig. 1), text-conditioned latent representations are transformed by the inverse flow and decoded into speech by the synthesizer. During training, posterior acoustic representations extracted from ground-truth speech are jointly optimized using reconstruction, duration, KL divergence, feature matching, and adversarial objectives. While this design provides multilingual synthesis and efficient speaker adaptation, it also couples acoustic representation learning with waveform generation, requiring the synthesizer to be jointly optimized with the remaining modules.

The waveform synthesis stage in MeloTTS employs the adversarial neural vocoder proposed in HiFi-GAN [14]. A fully convolutional generator is jointly optimized with discriminator networks to model waveform realism and periodic speech structure through adversarial and feature-matching objectives. This design has become standard in VITS-like architectures, improving perceptual quality beyond what mel-spectrogram reconstruction alone can achieve. However, because these vocoders operate directly in the time domain, they must implicitly learn waveform upsampling and phase-consistent reconstruction, tightly coupling the synthesizer to the overall optimization process and increasing both training complexity and computational cost.

VOCOS [7] offers an alternative formulation for the synthesizer module. Rather than directly generating waveform samples in the time domain, it predicts Fourier-domain coefficients from log-mel-spectrogram representations and reconstructs the waveform using an inverse short-time Fourier transform. This design maintains the single-pass efficiency of GAN-based vocoders while introducing an explicit time-frequency inductive bias into the reconstruction stage. Replacing the MeloTTS synthesizer with VOCOS therefore changes the role of the generator block. Waveform reconstruction is delegated to a pretrained frequency-domain synthesizer rather than learned through direct time-domain sample generation.

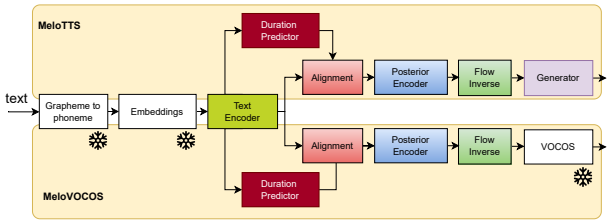


Figure 1: *Inference architectures: MeloTTS and MeloVOCOS. Frozen components are in a white background.*

3. MeloVOCOS

3.1. The proposed method

Fig. 1 compares MeloTTS and the proposed MeloVOCOS. Both architectures share the same linguistic front-end, text encoder, duration predictor, alignment mechanism, posterior encoder, and flow module. Given an input text sequence, the MeloTTS (top) converts text into phonemes and maps them to a latent acoustic representation. MeloVOCOS (bottom) replaces the HiFi-GAN-based waveform generator with a pretrained VOCOS decoder conditioned on log-mel spectrograms. The decoder is kept frozen, and the remaining MeloTTS modules are trained to predict VOCOS-compatible acoustic representations. This preserves the original linguistic, alignment, duration, posterior-encoding, and flow components while reducing the complexity of trainable waveform generation.

The posterior acoustic target is redefined to use log-mel coefficients as expected by VOCOS, rather than HiFi-GAN-learned features. Their computation is configured with the same FFT size (N_{FFT}), window length (N_{win}), and hop size (N_{hop}) as the pretrained VOCOS model. The posterior encoder must learn a latent representation that allows reconstruction losses to be computed directly in the log-mel domain.

Additionally, several adaptations were introduced to simplify optimization. The logarithmic transformation was applied before the normalizing flow, as applying it afterwards forced the latent space to approximate a pure linear spectrogram and degraded performance. An output projection layer was added after the inverse flow to better match the VOCOS input space. Several structures were evaluated, including convolutional, linear, and dense layers. We also report ablations for the discriminator, adversarial losses, and the flow module (last unreported).

Figs. 2 and 3 illustrate the practical differences between MeloTTS and MeloVOCOS. They show snapshots of the latent representation and the synthesizer input, respectively, with MeloTTS on the left and MeloVOCOS in the center. Fig. 3 also includes the ground-truth spectrogram on the right for the same utterance. The representations differ markedly. In our system, the input closely resembles the original mel-spectrogram, although the flow input is still far from a pure spectrogram. In MeloTTS, the HiFi-GAN input is instead clearly different and appears closer to the flow input.

3.2. Training MeloVOCOS

MeloVOCOS preserves the VITS-like objectives of MeloTTS that are independent of the original waveform generator. The frozen VOCOS decoder replaces the HiFi-GAN generator, while the discriminator networks and adversarial objectives are retained to encourage realistic speech after VOCOS synthesis. The training objective combines adversarial losses inherited from HiFi-GAN, the VITS duration (Dur), and KL-divergence.

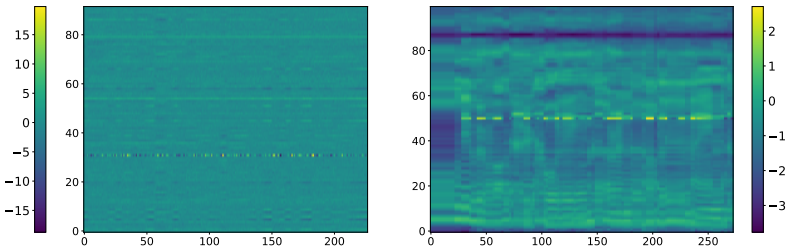


Figure 2: *Internal representations obtained by MeloTTS (left) and MeloVOCOS (right) before the normalizing flow.*

	M	F	Total
# utts.	16 K	19 K	35 K
# hrs.	14 h	17 h	31 h
Vocab.	46	46	46
Dur. (s)	$3. \pm 1.$	$3. \pm 1.$	$3. \pm 1.$

Table 1: *LibriVox Dataset Statistics.*

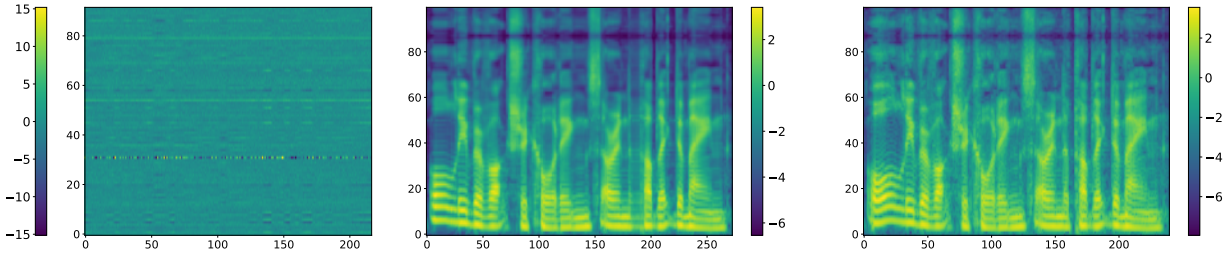


Figure 3: *Synthesizer input representations for MeloTTS (left) and MeloVOCOS (center), together with the ground-truth log-mel spectrogram (right). The x-axis represents frames, the y-axis represents coefficients, and the color scale is expressed in dB.*

We also introduced two new mel-reconstruction losses. The first supervises the predicted log-mel spectrogram, i.e. the VOCOS input. To account for the different power ranges across the mel-frequencies, we split the spectrogram into $K \geq 1$ subbands, each of which is weighted independently, preventing high-energy regions from dominating the reconstruction loss, $L_{\text{Mel}} = \mathbb{E}_{(s,\hat{s})}[\sum_{b=1}^{K} \alpha_b \, ||s_b - \hat{s}_b||_1]$, where s_b and $\hat{s}_b$ are the ground-truth and predicted log-mel representations in the b-th subband, and α_b is its weighting factor. The second loss evaluates the effect of the predicted representation after synthesis with VOCOS relative to the ground-truth mel-spectrogram: $L_{\text{Mel2}} = \mathbb{E}_{(y,\hat{s})}\left[||\phi(y) - \phi(G(\hat{s}))||_2^2\right]$, where y is the ground-truth waveform, $G(\hat{s})$ the waveform generated by the frozen VOCOS decoder, $\phi(\cdot)$ denotes mel-spectrogram extraction. Thus, L_{Mel} improves the intermediate log-mel reconstruction, while L_{Mel2} constrains the final acoustic output.

4. Experimental results

4.1. Dataset

We selected two speakers: female (F) and male (M), readers of English audiobook recordings from the LibriVox domain, Tab. 1. Recordings were preprocessed to obtain short clips that did not overflow buffers [15], resampled to 24 kHz, and converted to the WAV format. We used Whisper Large-v3 to produce transcriptions that matched the actual reading.

For each speaker, 10 utterances were used for adaptation out of the 10% training split. The validation (10%) and test (80%) subsets were used for checkpoint selection, convergence monitoring, and evaluation, respectively.

4.2. Preliminary assessment on VOCOS

Before integrating VOCOS into the proposed TTS pipeline, we assessed its robustness to imperfect log-mel spectrogram inputs. In our system, the acoustic model is not expected to produce ideal mel-spectrograms during inference, as prediction errors may accumulate, leading to distortion. To evaluate this effect, controlled perturbations were applied to ground-truth (GT) log-mel spectrograms extracted from the dataset, including:

- **Additive White Gaussian Noise** (AWGN), at different noise levels, to emulate prediction uncertainty and noise.
- **Frequency-domain smoothing** (FS), implemented as a one-dimensional moving-average filter with N coefficients along the frequency axis, to simulate spectral over-smoothing.
- **Time-frequency smoothing** (TFS), implemented as a two-dimensional moving-average filter with a $N \times N$ window over the temporal and frequency dimensions, to simulate broader spectro-temporal smoothing.

The utterances synthesized from the perturbed mel-spectrograms were evaluated using jitter, shimmer, HNR, PESQ, and ViSQOL. Lower jitter and shimmer indicate greater temporal and amplitude stability, while higher HNR, PESQ, and ViSQOL (VQ) indicate stronger harmonicity and lower perceptual degradation. Table 2 reports the results for AWGN at 20 dB SNR and FS/TFS filters with $N = 5$. VOCOS remains reasonably robust to moderate perturbations, but quality degrades when input loses spectral structure. This supports the need for log-mel features and reconstruction losses in MeloVOCOS.

C	S	Jit.↓	Shim.↓	HNR↑	PESQ↑	VQ↑
GT	M	3.1±0.5	1.4±0.2	1.0±1.3	4.3±0.2	4.6±0.1
GT	F	1.8±0.4	8.2±1.9	14.7±1.6	4.4±0.2	4.6±0.1
AWGN	M	1.4±1.3	7.3±2.2	7.3±3.2	3.5±0.3	3.9±0.2
AWGN	F	1.1±0.8	0.6±0.4	0.8±0.2	3.6±0.3	4.0±0.2
FS	M	5.6±1.6	2.7±0.9	1.3±0.1	3.1±0.3	3.5±0.2
FS	F	4.8±2.2	2.6±1.5	1.2±0.3	3.2±0.3	3.6±0.2
TFS	M	2.6±0.3	1.3±0.2	1.0±0.1	2.7±0.3	3.1±0.3
TFS	F	1.8±0.1	0.8±0.1	1.0±0.2	2.8±0.3	3.2±0.3

Table 2: *VOCOS robustness to perturbed log-mel inputs.*

Config.	CER (↓)	STOI (↑)	UTMOS (↑)	VISQOL (↑)	CosSim (↑)	FAD (↓)	Jitter (↓)	Shimmer (↓)	HNR (↑)
Baseline	1.0(0.0)	0.4(0.1)	1.0(0.1)	1.0(0.0)	0.0(0.0)	66.1(21.8)	0.1(0.0)	2.2(0.6)	17.7(2.6)
Conv	0.9(0.1)	0.3(0.1)	1.0(0.4)	1.0(0.0)	0.1(0.1)	79.6(19.7)	3.8(2.0)	11.0(2.6)	11.6(4.9)
Deep-C	0.8(0.1)	0.4(0.1)	0.8(0.1)	1.1(0.5)	0.1(0.1)	65.5(8.7)	4.2(0.6)	15.1(1.9)	9.8(1.8)
Deep-Cx2	0.9(0.1)	0.4(0.1)	0.9(0.4)	1.0(0.1)	0.1(0.1)	63.9(14.5)	4.6(0.7)	14.9(3.0)	8.1(3.3)
Deep-Tr	1.0(0.0)	0.2(0.3)	0.5(0.3)	1.0(0.1)	0.1(0.0)	49.7(13.3)	4.1(2.2)	24.7(7.0)	9.2(2.7)
Deep-FC	1.0(0.4)	0.3(0.1)	0.8(0.5)	1.0(0.0)	0.1(0.1)	94.4(24.2)	0.9(0.5)	8.8(3.0)	0.6(2.2)
G-Mel	0.9(0.1)	0.3(0.1)	1.0(0.3)	1.3(0.2)	0.2(0.1)	38.0(20.8)	3.8(0.5)	15.2(2.3)	10.9(1.2)
G-KL	0.8(0.0)	0.3(0.1)	0.8(0.0)	1.0(0.0)	0.1(0.0)	26.9(21.0)	6.4(3.5)	17.6(4.3)	9.7(3.1)
G-Dur	0.8(0.1)*	0.3(0.0)*	2.2(0.3)*	1.2(0.0)*	0.1(0.0)	33.8(17.9)	64.0(1.8)	11.3(1.9)*	13.0(1.8)*
G-Melx2	0.9(0.1)	0.3(0.1)	2.2(0.4)	1.1(0.0)	0.2(0.1)	35.3(13.6)	4.8(3.2)	12.1(3.6)	12.4(3.6)
G-Melx4	0.8(0.1)	0.3(0.1)	2.1(0.3)	1.1(0.0)	0.2(0.1)	35.8(15.3)	4.5(3.0)	11.9(2.8)	12.7(2.5)
A-Free	1.0(0.0)	0.2(0.0)	0.6(0.0)	1.0(0.0)	0.1(0.0)	209.8(52.2)	0.0(0.0)	0.6(0.1)	52.6(2.1)
256 ch.	0.9(0.1)	0.3(0.0)	2.0(0.3)	1.1(0.0)	0.1(0.0)	37.1(17.4)	4.4(2.3)	12.7(3.8)	12.5(2.8)
512 ch.	0.9(0.1)	0.4(0.0)	2.0(0.2)	1.1(0.0)	0.1(0.0)	45.2(15.5)	3.2(1.0)	10.7(1.3)	13.6(1.2)

Table 3: *Speech quality and audio metrics for evaluated training configurations. Best values within test block are marked with* *.

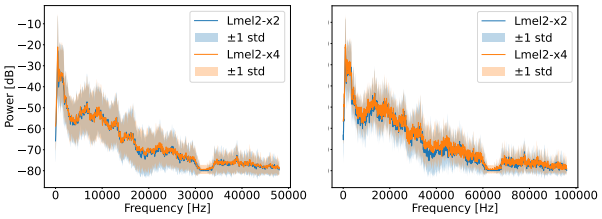


Figure 4: *STFT on G-Mel output frames @ ×2 (left), ×4 (right).*

4.3. Evaluated systems

We evaluated several configurations for integrating VOCOS into MeloTTS, listed in Tab. 4 and stating the (lin)ear or (log)arithmic mel-spectrogram, the projection layer used, and the losses used for the (G)enerator, the (D)iscriminator, and the (Dur)ation models. *Baseline* uses default MeloTTS. *Conv* inserts a convolutional projection between the flow output and the VOCOS input. *Deep* replaces the spectrogram representation with the VOCOS log-spectrogram target and evaluates several projection modules: linear, fully connected, and (multi)convolutional. *G* directly predicts VOCOS-compatible log-spectrogram features without projection and optimizes acoustic reconstruction losses. *A-Free* removes discriminators and adversarial losses, focusing training on duration and acoustic representation learning.

We also evaluated the proposed losses, $K = 4$. L_{Mel_2} complements L_{Mel} by adding a reconstruction constraint with wider temporal context. $L_{Mel_{x2}}$ uses FFT=WIN=2048 Sa. and HOP=512 Sa, for 10–ms temporal context. $L_{Mel_{x4}}$ uses FFT=WIN=4096 Sa. and HOP=1024 Sa, for 20–ms. Increasing temporal context from 10 to 20 ms reduces spectral variance and smooths acoustic representations between frames—Fig. 4 .

Computed audio metrics are reported in Tab. 3. The first column identifies the evaluated configuration and the additional losses used. The best configurations for the proposed architectures are marked. The results confirm the mismatch between the original MeloTTS representation and the VOCOS input space. Direct log-mel prediction with reconstruction-based objectives provides the best overall performance, while removing adversarial supervision degrades distributional quality.

Training efficiency is reported in Tab. 5, including wall-clock time, number of epochs, epoch duration, and active losses. MeloVOCOS reduces the total training time and number of epochs while keeping a similar per-epoch cost. The main gain comes from simpler optimization rather than faster iterations.

Name	Rep.	Proj.	G	D	Dur
Baseline	Lin	None	All	All	All
Conv	Lin	Conv	All	All	All
Deep	Log	Conv, Conv X2, linear, FC	All	All	All
G	Log	None	Mel, KL, Dur	All	All
A-Free	Log	None	Mel, KL, Dur	None	Dur

Table 4: *Summary of the evaluated training configurations.*

Decoder	Time (h)	Epochs	s/Epoch	Losses
HiFi-GAN	27.45	1260	78.4	All
VOCOS	6.63	300	79.6	Mel+KL+Dur

Table 5: *Efficiency of HiFi-GAN and VOCOS synthesizers.*

5. Ablation Study

The ablation study analyzes the main factors involved in replacing the HiFi-GAN generator with VOCOS, included in Tab. 3.

Acoustic representation. Figs. 2 and 3 show that the original MeloTTS latent representation differs substantially from the log-mel features expected by VOCOS. This confirms that VOCOS cannot be used as a direct drop-in replacement: the intermediate representation must be adapted.

Adversarial supervision. Adversarial supervision remains useful even when the waveform decoder is frozen. Removing the discriminator and adversarial losses degrades perceptual and distributional quality, indicating that discriminator feedback still regularizes the predicted acoustic representation.

Mel-reconstruction losses. The proposed losses improve this adaptation. L_{Mel} supervises the predicted log-mel spectrogram using frequency subbands, reducing the dominance of high-energy regions. L_{Mel2} adds a post-synthesis constraint extending the temporal context of the acoustic assessment.

Hidden-channel capacity. Increasing the hidden-channel dimension from 256 to 512 does not produce a consistent improvement across metrics. This suggests that MeloVOCOS is limited more by representation compatibility and loss design than by latent capacity.

6. Conclusion

We proposed MeloVOCOS, a MeloTTS-derived architecture that replaces the original HiFi-GAN with a frozen VOCOS synthesizer. The results show that waveform reconstruction can be partially delegated to a pretrained synthesizer while preserving intelligibility, speaker similarity, and competitive speech quality with reduced training cost. Ablations indicate that successful integration of VOCOS requires explicit log-mel adaptation, reconstruction losses aligned with the VOCOS input space, and adversarial feedback to produce high-quality synthetic speech.

Future work will explore multi-synthesizer training schemes that combine different synthesis flows and reconstruction objectives to improve robustness, quality, and adaptation across speakers and languages.

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